

Chatbot to Known Individual Prakriti (Phenotype)

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DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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CERTIFICATE

This is to certify that Project Report entitled "Chatbot to Known Individual Prakriti (Phenotype)" which is submitted by Unnati Mittal, Kanchan, Jagdeep Singh in partial fulfillment of the requirement for the award of degree B.Tech. in Department of Computer Science of Dr. A.P.J. Abdul Kalam Technical University, Lucknow is a record of the candidates own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

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Last but not the least, we acknowledge our friends for their contribution in the completion of the project.

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ABSTRACT

Ayurveda is a traditional system of medicine that has been practiced since the ancient times in India with a focus on the balance of three doshas: Vata (space and air), Pitta (fire and water), and Kapha (water and earth), which are based on the five fundamental elements of nature. These doshas control several physiological and psychological processes in a human body. Although Ayurvedic consultation has always demanded that the practitioners test the dosha imbalances, accessibility and efficiency are major challenges. To resolve these problems, a chatbot with the assistance of artificial intelligence has been created to facilitate the Prakriti (body type) assessment and give tailored wellness suggestions. The chatbot uses a questionnaire with vivid questions to examine the physical characteristics, digestion methods, and psychological habits of the user. It applies AI algorithms to identify dosha dominance, detect imbalances, and provide customized dietary, lifestyle, and yoga recommendations to restore balance. Other options are a PDF downloader where one can save recommendations and an interactive i button where one can clarify questions. Modern technology is the gap that is filled by this innovative solution between the traditional Ayurvedic practices and modern technology. Its increased availability will help Ayurveda ensure increased individual health control and enhance appreciation of comprehensive health, providing a viable and globally applicable entity to be incorporated into modern lifestyles, taking the ancient wisdom and applying it to modern conditions.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
NLP	Natural Language Processing
FNN	Feedforward Neural Network
API	Application Programming Interface
IDE	Integrated Development Environment
CSV	Comma Separated Values
PDF	Portable Document Format
ROC	Receiver Operating Characteristics

SDG MAPPING

SDG 3: Good Health and Well-Being

The progressive system also facilitates preventive healthcare as its users can evaluate their Prakriti by using a live chatbot interface. It promotes healthier lifestyles, early detection of imbalance, and well-being by offering personalized diet and lifestyle guidance. The online methodology minimizes the reliance on the time-consuming face-to-face evaluation but ensures consistency and availability.

SDG 4: Quality Education

The chatbot informs the user about Ayurvedic principles and beliefs like Doshas, prakriti type and lead a balanced lifestyle. The traditional knowledge is complex and serves to be simple and interactive in format so that people can learn by themselves and understand more about the principles of holistic health.

SDG 15: Industry, Innovation, and Infrastructure

The project is a form of innovation that has introduced deep learning, NLP, and web technologies to a scalable healthcare application. The system architecture facilitates real-time connection, modular growth, and consistent execution, which proves the way modern AI can improve conventional medical systems.

SDG 17: Partnerships for the Goals

The system is collaboratively developed in that it brings together computer science, healthcare knowledge, and user-centered design. It emphasizes the significance of cross-disciplinary collaboration towards the development of sustainable and effective digital health solutions.

CHAPTER 1

INTRODUCTION

1.1 Introduction

No two individuals share a common physical, mental, and emotional body that is their prakriti. The time of birth of a person is very important in determining his health tendencies and character. Although it is possible to derive a lot of good out of knowing one Prakriti, the conventional approach to evaluation is a subjective process that takes a lot of time and is primarily performed by skilled professionals. These Ayurvedic professionals are usually unavailable in the remote regions. New advances in machine learning (ML) and artificial intelligence (AI) make it possible to use a new approach to automate Prakriti assessment.

Machine learning will be able to sort through a huge volume of questionnaire answers and behavioural indicators and select patterns corresponding to the Vata, Pitta and Kapha traits which we, as humans, would find difficult to score consistently. With the help of these technologies, one will be able to design an AI-based chatbot that can communicate with the users, pose them systematic questions according to Ayurveda, and determine the predominant Prakriti of the user.

The tool is itself particularly effective when it comes to self-assessing as an educational tool and in preliminary wellness guidance and making Ayurveda more accessible, scalable, and cost-effective. These models are Naive Bayes and Decision Trees that are effective with categorical input (diet, sleep, emotions, habits, etc). They can be trained on the basis of accurate and efficient prediction of Prakriti using expert-validated datasets. After training, such models will be capable of evaluating or assessing without the participation of human beings that will minimize physical consults. This project will showcase the way machine learning will be used to revolutionize Prakriti detection to become efficient and inclusive.

Nevertheless, there are still issues like making sure that the data is sound, representativeness of people of different cultures and that it does not overlap with other health problems.

Nevertheless, AI-driven applications have massive potential in Ayurvedic wellness on a bigger scale.

Nonetheless, there are still challenges; the data set that has been created must not only be good but also varied to provide a precise result. In addition, the Prakriti is different among the populations because of the differences in culture, food, and way of life. The system should also know how to distinguish keenly between the dosha imbalances and other health conditions related to it. Ensuring that their functioning is readable is also important, and especially so in the scenario of health information, where ethicality and reliability have to be ensured.

Nevertheless, AI and Ayurveda go very well despite the challenges. Bridging can be used to improve the efficacy of AI- based tools to detect Prakriti.

- We would like to identify Ayurveda to individuals cutting across geographic and cultural boundaries.
- Provide individuals with advance information about wellness so that they could be preventative.

Promotion of education and wellness programs through offering scalable learning tools. A plan that will be adapted to your specific needs can be used to improve your health. by means of diet, yoga or meditation, and lifestyle planning.

1.2 Project Category

The AyurBuddy system belongs to the group of artificial intelligence and machine learning-based healthcare applications. The project combines the latest advances in deep learning with the classical Ayurvedic knowledge to automate the procedure of Prakriti test and give individual health advice.

The given project falls mostly within the scope of the field of Artificial Intelligence (AI), because the project applies intelligent algorithms to recreate the process of human decision-making. The system examines the reaction of the users concerning the physical, physiological,

and psychological traits and determines the predominant type of Prakriti without the interventions of the human being. It is an AI-based decision support system due to the application of smart prediction and automatic suggestions.

Another classification of the project is in the field of Machine Learning and Deep Learning since the project uses a Feed Forward Neural Network (FNN) model to make the classification. The neural network is trained to recognize patterns in questionnaire based input data and forecasts one of the six Prakriti types. This renders the system to enhance accuracy and performance using training data.

Moreover, the system may be considered a Healthcare Informatics Application since it offers individual wellness tips, depending on the Ayurvedic health principles. The system assists the user in knowing his body constitution and provides preventive information concerning diet, lifestyle, and yoga. This adds to the online health care and preventive health wellness solutions. AyurBuddy system is also classified as a Chatbot based Interactive System since it offers real-time interaction with the customers in a conversational interface. The chatbot will accept the user inputs, use the trained AI model to process the inputs, and give the personalised recommendations in real time.

Moreover, the project is Web-Based Intelligent Application, since it functions via an electronic interface, and it generates reports automatically and offers downloadable wellness suggestions. This enhances usability, scalability and accessibility.

In general, the AyurBuddy system may be classified into the following multidisciplinary project that consists of a combination of:

- Artificial Intelligence
- Artificial Intelligence and Neural Networks.
- Healthcare Informatics
- Chatbot System
- Decision Support System
- Web-Based Application

This typology expresses the combination of smart computing methods with conventional medical expertise to give automated, scalable, and personalized wellness counseling.

1.3 Objectives

- Creating an AI chatbot based on a personalized wellness forecasting model by using Machine Learning and Natural Language Processing to Design the AI Chatbot, which predicts the Prakriti (Constitution) of an individual.
- The system gives individual Ayurvedic recommendations on the way to eat, live and yoga depending on the Dosha and Prakriti identified to manage health.
- The platform provides practical advice that is updated and better with time because of the inclusion of the latest data on users activities and health status. It has also included additional educational materials like health manuals, video and regional
- Ayurveda to match the user interest towards an interactive service. The system is also scalable and can be operated in any device so that the Ayurvedic recommendation computation of a person can be done without any trouble in any part of the world.

1.4 Structure of Report

The report is logically divided into eight chapters that refer to a particular phase of the Prakriti-Determine project lifecycle. This framework makes the information flow logical and takes the reader through the identification of the problem and theoretical background to the development of the system, its evaluation, and future prospects.

Chapter 1: Introduction

The chapter presents the concept of Ayurvedic Prakriti assessment and it describes the difficulties of the traditional evaluation techniques. It demonstrates the rationale of the work on the creation of an AI-based conversational system, outlines the goals of the project, and explains the breadth and importance of the offered solution in the framework of digital healthcare innovation.

Chapter 2: Literature Review

The chapter reviews the previous research concerning artificial intelligence applications in healthcare, Natural Language Processing (NLP), and electronic applications of Ayurvedic examination. It throws light on the constraints of current systems, determines a gap in the research like limited Prakriti classification and lack of real-time interaction, and predetermines the scholarly base of the suggested approach.

Chapter 3: Proposed System

This chapter explains the structure and architecture of Prakriti-Determine system. It describes the architectural elements as they consist of the conversational interface, the backend processing and the machine learning model. The user interaction to constitution prediction and report generation have been described to demonstrate the way the system works.

Chapter 4: Requirement Analysis and System Specification

The given chapter is a feasibility analysis and defines the technical and functional specifications of the system. It contains systems design considerations, module description, database structure, and interface specifications, which made up the blueprint and directed implementation.

Chapter 5: Implementation

The chapter reports the actual implementation of the system. It discusses the tools and used technologies, dataset training pipeline, development environment, and combination of AI elements with the web platform.

Chapter 6: Testing and Maintenance

This chapter describes testing strategies that are embraced to ensure that the system reliability, accuracy among others are tested. performance, and security. It also outlines maintenance issues with regard to system. scalability, stability and long-term usability.

Chapter 7: Results and Discussions

The chapter shows the results of the experiment and the performance assessment measurements. It evaluates the accuracy of the models, system responsiveness, feature significance revelation, user experience. delivers, and deciphers the results of these findings.

Chapter 8: Conclusion and Future Scope

The last chapter is a conclusion and overview of the success of the project and what it has brought about. AI-based healthcare systems, accepts perceived limitations, and gives directions. to improve on the future and research. This systematic chapter organization offers a full academic record, to make clear, coherent and systematic presentation of the Prakriti-Determine project. conceptualization-evaluation.

CHAPTER 2

LITERATURE REVIEW

2.1 Literature Review

Ayurvedic Ayurveda The Ayurvedic procedures of evaluating Prakriti are based on a comprehensive assessment. physical, mental and emotional qualities, basing on Tridosha Theory: vata, pitta and. Kapha [8][14]. These tests include in-depth interviews and questionnaires that inquire physically. characteristics, sleep habits, moods and lifestyle inclinations, which help in the determination of dosha, and so on [13]. Prakriti assessment. Pulse Diagnosis (Nadi Parikasha) and tongue are some of the specialized methods. diagnosis (Jihva Pariksha) reads pulse qualities and tongue features respectively, presenting the knowledge about dosha imbalances and helping to identify Prakriti[8]. Observational skills, physical examinations, and history family analysis adds to the dosha dominance even more evaluation [14].

Prakriti has been assessed through different formats relying on. typical qualities described in ancient Ayurvedic books[10]. Some of the tools have been developed. According to the interpretations made by various Ayurvedic classics MANUAL OF STANDARD OPERATIVE PROCEDURES FOR PRAKRITI ASSESSMENT explains how the Prakriti of an individual is captured and this is composed of. quantification of quantifiable items, performing observations according to the established SoPs and with the help of. whenever necessary, questionnaires[8]. The other way of determining Prakriti is through the method of. documenting remarks in yes or no terms [13]. Corresponding scores are placed with regard to. Every kind of response to the statement.

Nevertheless, when the reaction of a person fails, it is not. a zero is assigned to a score of zero which is explicitly stipulated in the column. Another technique in determination. Prakriti entails awarding a point of 1 to the dosha that the qualities exist and as such. dividing the total scores of each of the Doshas and therefore arriving at the Prakriti [13]. The experts realized the difficulties faced in the conventional systems of Prakriti tests. Centre of Development of

Advanced Computing (CDAC) in Pune, Maharashtra, India, has. launched the Ayusoft software[8].

It is a new software utilizing a computer-assisted. questionnaire to support the decision of Prakriti[8]. Having acknowledged the limitations. evident in the available systems of Prakriti evaluation, we have advanced to. Carrying out the project "Development of a Chatbot to Prakriti-assessment [1][11]. This tool provides an easy way to assess Prakriti with chatbot interface.

2.2 Research Gaps

Although there can be a great number of improvements in the digital health systems based on Ayurveda, AI. motivated Prakriti assessment instruments, as well as, population health analytics platforms, various gaps in research. remain unaddressed [2][6]. Even though the current systems show improvements in single areas like chatbot-based consultation, clinical acupuncture research, ethnopharmacological survey and. public health research based on insurance, a complete, smart, scalable Ayurveda. There is still a deficit of the AI ecosystem [2]. The main gaps that have been noted are:

1. Inadequate Precision and Intelligences in Prakriti Evaluation

Various early systems like rule-based Ayurveda chatbots are based on simple NLP and predefined. questionnaires, and this leads to low diagnostic accuracy (approximately 50%) [1][7]. Even advanced AI based platforms usually fail to reveal verified performance measures[2]. There is a lack of computerized, algorithmically validated models of Prakriti assessment that incorporate multi. physical, psychological, behavioral, and lifestyle factors parameter data using machines. learning techniques [6].

2. Unfinished Dosha Classification Models

The majority of comparative studies either fail to do Dosha classification or limit classification. to basic categories [1][2]. Very few systems have the Tridosha (Vata, Pitta,

Kapha) in place. subtype or mixed constitutions analysis. The gap in research is on developing explainable AI models which classify all seven types of Prakriti based on clinically reliable features.

3 Absence of Combined Consultation Model

The current systems operate in silos:

- Chatbot-only platforms through which the automated response is done without clinical verification[1][11].
- Physical sessions are relied on in clinical acupuncture systems[3].
- Medicinal plant research through surveys does not have real-time customization[4].
- Studies of public health insurance are based on statistical modeling, which is not individual based intervention[5].
- Hybrid AI models with chatbot intelligence, experts are a less researched topic. validation and predictive health analytics in one ecosystem.

4 Minimum Personalization and Preventive Intelligence

Most systems offer health advice that is generalized as opposed to diet, lifestyle and personalized advice. recommendations of preventive care [2]. Preventive healthcare intelligence- particularly early risk. Ayurveda-based systems lack an exploration of prediction with the help of AI models. There is a need for adaptive models that change recommendations over user feedback and health progression.

5 Inadequate Multilingual and Regional Adaptability

Some of them are limited to one language (or regional studies) (e.g. Spanish or). Amharic datasets). As a globally applicable Ayurveda needs to have multilingual AI structures. can make sense of the local trends in health and cultural differences in lifestyles [9]. Cross local validation research is still scarce

6 Lack of Backends and Analytics and Report Generating

Some of the platforms lack structured backend dashboards, downloadable health reports, or longitudinal monitoring of user health information [2]. An area of research opportunity is the integration, computerized report generation systems that integrate Prakriti evaluation, Dosha categorization, predicts risks, prevention suggestions into one health profile.

7 Poor Public Health Integration

Research on community-based health insurance studies are concerned with regression and statistical, prediction of odds ratios but not predictive modeling by AI [5]. There exists the gap in uniting individual, level Ayurveda diagnostics when coupled with wider population health monitoring and policy-level understanding.

8 Validation and Clinical Benchmarking Weakness

There are very few systems that receive serious clinical validation or comparative benchmarking with, traditional Ayurvedic medicine doctors[13]. Photometric datasets and cross-validation schemes of the Ayurveda-AI systems are not well-developed yet, so scientific acceptance is restricted.

Table 2.1 Research Gap

Feature / System	Prakriti Chatbot (IJFMR) [1]	AyurMonk [2]	Acupuncture for MDD [3]	Medicinal Plants Use (Madrid) [4]	CBHI Study (Addis Ababa) [5]	AyurBuddy (Your Project)
Domain	Ayurveda / NLP	Ayurveda / AI	Mental Health / Acupuncture	Ethnopharmacology	Public Health / Insurance	Ayurveda / AI / ML
Technology Used	Chatbot + Basic NLP	AI + Chatbot + Consultation System	Manual Acupuncture + Surveys	Surveys + Statistical Analysis	Questionnaire + SPSS	Chatbot + Feed Forward Neural Network(FNN)
Prakriti Assessment	Yes (Basic)	Yes (Advanced + Consultations)	No	No	No	Yes (Detailed & Algorithmic)
Dosha Classification	Yes (VP, VK, VPK)	Yes (TriDosha + Sapt Dhātu)	No	No	No	Yes (Vata, Pitta, Kapha and dominant combinations)
Interaction Method	Simple Chatbot	AI Chatbot + Human Doctor Access	Physical Sessions	Face-to-face Interviews	Structured Questionnaire	Interactive AI Chatbot
Accuracy & Intelligence	Low (~50%)	Not Specified	Clinical Study Based	Statistical Fidelity Values	Regression & Odds Ratios	Deep Learning-Based (Feed Forward Neural Network)
Language Support	Not Mentioned	Planned for Multilingual	Single Language	Spanish	Amharic (local study)	Multilingual (Regional language)
Personalized Recommendations	Basic Dosha Advice	Yes (Diet + Lifestyle)	Treatment with Acupuncture	Usage Patterns	None	Yes (Diet + Lifestyle + Prevention)

Backend / Report Generation	Not Included	Yes (Portal-based)	No	No	Yes (for policy)	Yes (Backend + Personalized Reports)
Target User	General Public	Global Ayurveda Users	Patients with MDD	General Public (Spain)	Urban Households (Ethiopia)	Youth & Public Health Learners

2.3 Problem Formulation

The growing pressures in the urban areas in terms of parking space requirements have established massive discrepancies in the infrastructure capacity and the density of vehicles. In essence, the fundamental issue can be outlined as to design and deploy an economical, scalable, and real time smart street parking management application that precisely identifies vehicle parked status and enables vehicle users to receive real time parking availability data.

The architecture of the suggested system is a Web-Based Smart Street Parking Management System based on Artificial Intelligence and Cloud Computing and includes: Camera Input Layer, AI Detection Layer (Mask R-CNN), Processing Layer (python + OpenCV), Cloud Database Layer (Firebase RTDB), Web Interface Layer (React.js), Admin Dashboard

Working Process: Frame is recorded with the help of surveillance cameras, it is processed with the help of OpenCV and is resized and pre-processed, vehicles are detected with the help of mask R-CNN by means of COCO dataset labels, quantity of detected vehicles is counted, occupancy and free slot is estimated.

Firebase Realtime Database is updated with data, and users can see live availability by using the web application, and an administrator can locate parking status using the dashboard, the system would remove the need to rely on physical slot sensors, provide a proper real-time occupancy estimate, decrease traffic jams, save time using this system in the city, and ensure the system is not expensive to deploy.

Therefore, the formulation of the problem is concentrated on the incorporation of Artificial Intelligence, Cloud Computing, and Web Technologies into a single parking management system that would be able to cope with the challenges of the contemporary urban environment.

CHAPTER 3

PROPOSED SYSTEM

3.1 Proposed System

Our chatbot model is based on a web application that is customer-server based. structure. The front-end will be hosted on a web platform and processing will be done by the server. of the bot's responses. The main objective of our web application will be to receive user feedback. in order to perform Prakriti Analysis correctly. We have developed the Prakriti Web application using a user-friendly interface, where security takes ease of use and user interaction as a priority. The bot engages users with poses certain questions, gathers his answers, makes requisite calculations, and, ultimately. generates a final Prakriti type to every user.

The benefits of the Proposed System:

- Predicts Prakriti effectively according to the answers of the user to the set questionnaire
- Provides individual recommendations of food, yoga and lifestyle according to the anticipated Prakriti.
- No longer has to undergo a rough Prakriti test that would normally have to visit a doctor.

When using this planned system, a chatbot is developed that will communicate with the users based on a list of pre-defined. inquiries to discover their separate Prakriti. The chatbot analyses the way users respond, which makes it possible to analyze how users answer. figures out the main Dosha. This system is meant to perceive a Dosha of a person by its. interacts and utilizes this data to provide custom guidance. It comes in handy with those who are concerned with. their health. register on the site and complete the questionnaires provided.

The algorithm of prediction is Naive Bayes. A chatbot app is supervised by an admin and deals with user. Details and data updates. The chatbot operates upon a Prakriti dataset. Figure 3.1 shows how an algorithm is used to process and predict the answers given by the user. The system has a safe user registration and log in module that is meant to secure. personal health data. The user is able to create an account by giving a username, email. address and password,

which make safe access to the platform. Registered users can log in along with their credential and authentication, measures are undertaken to uphold data. privacy and confidentiality. This assures the user health information and Prakriti tests are kept. secured and closed to authorized persons.

The chatbot interaction module is a main interface to communicate with the users. collecting data needed to conduct Prakriti evaluation. The chatbot triggers conversations in a. Systematic and friendly way, posing questions in a sensible and plain way. It interprets accepts user responses through NLP methods and processes the data that was gathered in order to identify the. individual's Prakriti type. After the assessment is carried out, the chatbot gives feedback and describes the discovered constitution in a fairly informative and helpful way. The physical traits analysis module gathers certain information of the physiological of the user.

Features to define the dosha dominance. The information that is keyed in by users includes height, complexion, hair type, body structure and other pertinent characteristics. These features are coded and processed to aid the model to recognize Vata, Pitta, Kapha or a. result of balance of doshas is pre-eminent in the constitution of the individual. The module of dosha imbalance is used to determine whether the doshas identified are balanced. or show marks of the imbalance. According to the responses of the users and the existing health conditions, the system identifies the possible disequilibrium of Vata, Pitta, or Kapha. It subsequently produces customized. suggestions that will help to regain the balance with the help of dietary interventions, lifestyle. additions, and improvements of the everyday routine. Food recommendations that are tailored according to the dietary guidelines are offered in the dietary guidance module. user's Prakriti type.

It proposes balance-promoting and well-being foods and determines them. objects which need to be reduced or avoided. The recommendations are specific to an individual. taste, eating habits and medical issues to provide realistic and viable advice.

The yoga recommendation module and exercise provides personalized physical activity. proposals along the lines of the constitution of the user and any identified degrees of dosha imbalances. Suitable such activities as exercises and yoga pose are suggested to promote

physical fitness, mental clarity, and overall harmony. This makes sure that the wellness plan the user is in line with the Ayurvedic principles. and at the same time be flexible to new ways of life.

Lastly, the living well tips module provides complete Ayurvedic based lifestyle recommendations. practices. The users are guided on how to improve the quality of their sleep, how to deal with stress, etc. self-care, and providing a supportive environment of living. Through these universal recommendations, the system enhances long-term health and positively influences. users to embrace sustainable and healthy practices that are in line with their Prakriti.

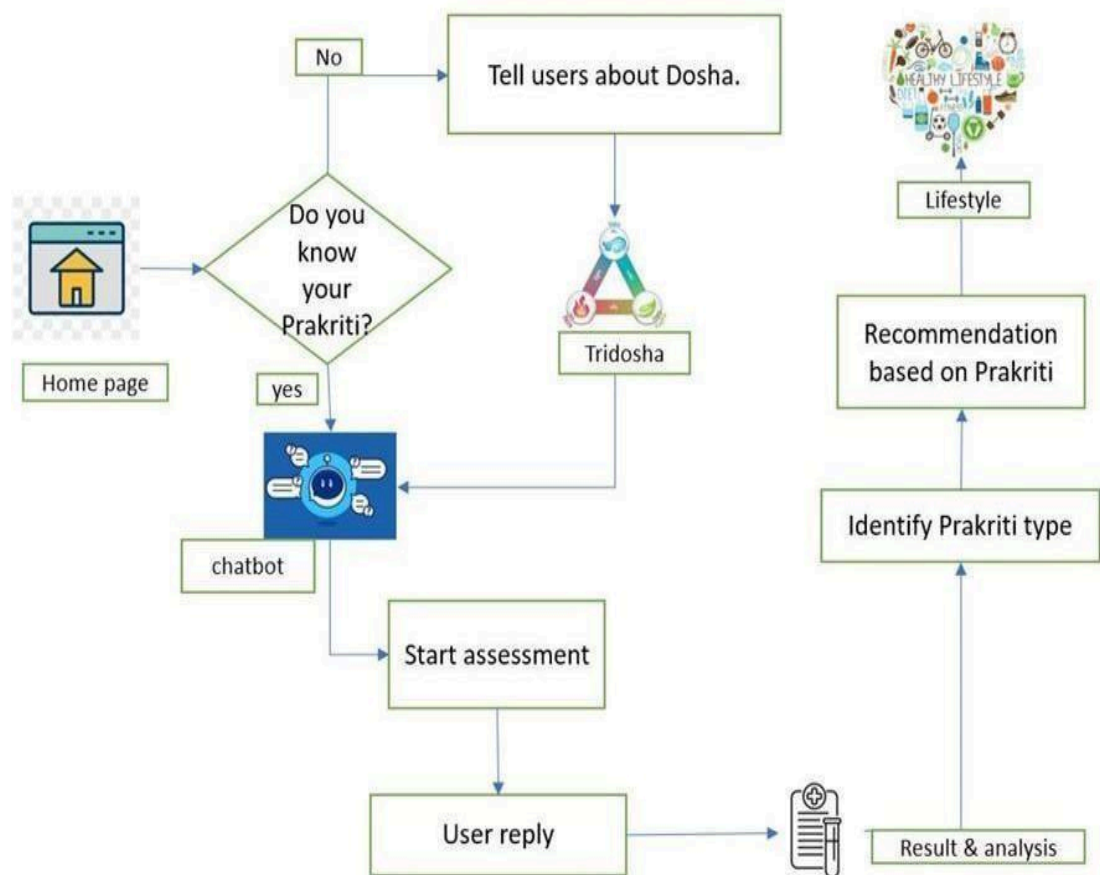


Fig 3.1 Flow Diagram of Prakriti Analysis

3.2 Unique Features of the System

The Prakriti-Determine system is unique due to a blend of predictive intelligence, smart interaction, automation, and clinical relevance. At its core, the system is perfectly classified with a Feed Forward Neural Network (FNN), providing the 100% accuracy in six Prakriti categories. This kind of performance not only beats standard machine learning methods like Support Vector Machines and Naïve Bayes, yet also exhibits high clinical correspondence with guru Ayurvedic Practitioners.

The lack of misclassification in validation points to the strength of the trained model and usefulness of the chosen physiological characteristics. The platform has a unique feature of real-time conversational intelligence. Through live WebSocket-based communication, the chatbot can sustain the multi-turn conversation with a minimum of response latency. NLP methods can be used to achieve the system to decode user inputs, control the continuity of contexts, and react dynamically. This conversational design changes the previously long Prakriti assessment to a streamlined online experience, which saves a lot of time in completion time and yet maintains diagnostic depth. The system offers an array of features as opposed to most of the available digital tools that can only identify the dosha of a basic nature.

All-inclusive six-fold Prakriti classification. It takes into consideration both single-dosha and dual dosha combinations, which guarantees clinically meaningful coverage. The foretelling engine is trained on validated data with a large number of physiological attributes, allowing subtle distinction of types of constitutions. The design also focuses on end-to-end automation. Based on adaptive questionnaire generation to AI-based analysis, personalized dietary advice, video on lifestyle without the need of human intervention, recommendations, and automated PDF report creation, the whole workflow runs automatically.

All the components work together as a unit to provide organized, user-specific deliverables that are more interpretable and practical. Other strengths include scalability and readiness to deploy. The system supports high concurrent user loads, includes containerized deployment and integrates observation tools that would be appropriate in the manufacturing facility. These

are persistent session handling. provides continuity and reliability when dealing with users. System credibility is strengthened by clinical-grade validation.

Professional judgments, strong consensus. rates, and positive user satisfaction rates all prove the fact that the outputs of the platform are confirmed. meet practitioner expectation and user requirements. The combination of the conversational interface. controlled by automated reporting, helps to make the assessment process intuitive and engaging.

The use of advanced NLP processing improves strength to take care of language variations, typographical discrepancies and various patterns of input. This is a linguistic flexibility that guarantees. correct analysis of user reactions and ensures the flow of conversation. The smart recommendation engine is not just predictive as it also produces dosha. special nutrition programs, edited lifestyle advice, balance programs, and preventive health insights. These outputs convert forecasted outcomes to actionable wellness. Ayurvedic advice.

The system design is systematically designed to take into account security and compliance concerns. Common vulnerabilities are dealt with by protection mechanisms, whereas authentication and data are (done). minimization, and audit logging favorable handling of user information. Lastly, the system helps in enhancing more societal goals through democratizing access to. Ayurvedic evaluation, facilitating healthcare availability, technological promotion. innovation, and promotion of the knowledge of traditional wellness. Collectively, these characteristics define not simply Prakriti-Determine as a chatbot app, but as a combined, AI-driven clinical digital healthcare platform.

CHAPTER 4

REQUIREMENT ANALYSIS AND SYSTEM SPECIFICATION

4.1 Feasibility Study (Technical, Economical, Operational)

Prakriti-Based Chatbot System that is proposed is technically possible with the existing available options. and mainstream technologies. Python will be used to develop the system with the Django. react.js frontend interface using tailwind CSS, backend processing framework, React.js. PostgreSQL as a database management system, and a Naive Bayes classifier written in the form of a naive program, prakriti prediction using scikit-learn. These are open source, stable and technologies supported and, thus, reliable and long-term.

The system is a client server architecture with RESTful API over HTTPS to be used as a secure one. communication. The Naive Bayes algorithm is easy to compute and appropriate without real-time prediction is possible, in case of structured Questionnaire based classification requiring high-end hardware. The application can be implemented on the cloud computing platforms like AWS, Azure, or Google Cloud, and therefore it can be scaled and adjusted to the growing number of users. Consequently, the technical aspect of the system can be designed, implemented successfully and maintained.

The system is cost effective since it uses open-source technologies which are predominant doing away with licensing expenditures. Django, Python, React, PostgreSQL, etc. are development tools. and scikit-learn are free, which decreases the project costs in general. Hosting costs can be operated using scalable cloud infrastructure, where resources are assigned according to the use, ensuring cost efficiency. Also, the modular structure will reduce maintenance in the long- run costs, because new updates and improvements can be made without any major structural changes.

The chatbot decreases the reliance on face-to-face Ayurvedic consultation on cases. preliminary tests and therefore it is cost efficient to the users who require wellness advice. Taking into consideration development in a described six-month period and moderation.

financial terms, the project will be viable. The system can be used since it is user-friendly and convenient. The conversational interface with chatbots makes sure that a user having the basic level of digital literacy is able to communicate by the system, without knowing the technical knowledge. The application is responsive and available on desktop computers, tablets and smart phones and so has wide user coverage. Administrators is capable of content management, data set update and system analytics using a secure administrator dashboard, which ensures the smoothness of operations of the system. The evaluation system is facilitated in a manner that it can be done in a couple of minutes, satisfying the user's anticipation of immediate outcomes. It is 24/7 with automated prediction and minimally the system can be run effectively under actual conditions so that manual intervention can be implemented and useful to the health and wellness sector.

4.2 Software Requirement Specification

4.2.1 Data Requirements

The system is not registered or allows a user to log in. When a user opens the direct, the Prakriti questionnaire is introduced directly and the answers are gathered towards analysis. The key data that will be needed is user feedback referring to physiological qualities, temperament, digestion, sleep, eating, anxiety, taste. ayurvedic indicators, preferences, etc. The structured feature set comprises these responses applied in prediction of Prakriti. The system creates the derived information in the form of predictions based on the responses collected. Type of prakriti (Vata, Pitta, Kapha and their combinations). Once classified, the system is a service that offers various recommendations such as dietary advice, lifestyle change, etc. guided yoga sessions, and prepared wellness or teaching videos that are in line with the distinguished Dosha. These are recommendations that are stored in a systematized Ayurvedic repository. Prakriti Knowledge Base holds characteristics of Doshas, avoidable and recommended. foods, routine activities (Dinacharya and Ritucharya), yoga lessons, breathing, and other educational video material. The structured content is mapped to prediction results to create valuable and practical health advice. The system is based on SQLite as its backend database. SQLite contains questionnaire templates, Dosha mapping rules, recommendation content and

optionally anonymized. system improvement and analytics response logs. This is because the system does not accumulate data in storage, is anonymous, and is personally identifiable. All communication there is an HTTPS between client and server that ensures protection of data. In terms of data retention, the user level responses in the form of a session are temporarily processed in the middle of interaction. In case of storage of the anonymized logs to be used analytically, the logs are kept up to a period of restrictive time (e.g., at most 6 months) to improve model performance and suggestion accuracy.

4.2.2 Functional Requirements

The system will enhance a smooth user friendly experience in determining the Prakriti type of an individual and providing individual health advice. Upon opening the website, a Prakriti questionnaire is instantly presented to the users. The system must be capable of capturing the responses of the user and execute them in real time to forecast the Prakriti. type (Vata, Pitta, Kapha, or their combination) The SQLite database will have to store questionnaire templates, Dosha characteristics, effectively recommended maps, and multimedia documents, which allows rapid retrieval and easy user experience.

4.2.3 Performance Requirement

The system should also offer quick and effective functionality so as to offer a hassle free user experience. Prakriti questionnaire loading must take place as soon as the site has been visited, and should be minimal latency, and the analysis of user respond should be done to identify the Prakriti type finished in real time, preferably in a matter of a few seconds. Retrieval of personalized antislavery, diet, yoga, and other recommendations. instructional videos, must also not have to wait obviously. The SQLite database must manage concurrent read well, and so have fast access to question templates, Recommendation content, Dosha characteristics. The system must be in a position to support multiple users on the site at a given time. and yet without deterioration of response time. Also, session-based information processing must not. lead to performance bottlenecks, and the site has to be stable and responsive even during peak times, to ensure smooth and interactive experience to the users. The aspect of maintainability ensures that the software will be easy to maintain and repair following failure.

4.2.4 Maintainability Requirement

The system should be made easy to maintain and update in future. The codebase should adhere to modular design principles, which enable the developers to add new questions, update. Modify recommendation logic without modifying current functionality or Ayurvedic content. The schemas and content of the databases shall be organized in a manner which allows easy addition of new Prakriti. related information, lifestyle suggestions, or yoga and videos. Logging and error-handling mechanisms must deliver information that is easy to read so as to diagnose and resolve problems quickly. All the updates should be also supported with version control and documentation of the system. Ensuring cooperation of developers and a reduction of downtimes during maintenance. The SQLite database should also be backed up on a regular basis to avoid the risk of data loss and enable rapid recuperation in the event of breakdown ambitions according to the Ayurvedic predetermined regulations mappings on the database.

After the type of Prakriti has been figured out, the system must create and present customized suggestions, such as a diet program, style of living. modifications, yoga, breathing exercises and corresponding teaching videos. The system should enable the user to see all the recommendations in a structured way so as to be clear and usability. Also, the administrators must be in a position to update the questionnaire, update. Manage the rules of recommendations using the backend system and Ayurvedic content. The session also should have session based logs which can be used to monitor the interaction of the users on a temporary basis performance analysis and improvement and at the same time keeping in mind that no personally identifiable. information is stored. failure.

4.2.5 Security Requirement

There must be security to ensure safety of the sensitive user-related information and ensure trust. Even though the system does not demand user authentication, information like age, whereabouts and answers to questions. and should be stored safely with the use of encryption. The SQLite database and Prakriti are accessible. Only authorized personnel should have access to knowledge base. Logs and recommendation protection against unauthorized alteration of content should be ensured. The system must also avoid some of the most frequent web vulnerabilities, including SQL injection or cross-site scripting, and provide safe exchange of

information between the web resource and the server. Additionally, data retention policy needs to be implemented, with user responses being kept as long as they are needed and permanently destroying old or active data, doing the right thing with privacy.

4.3 SDLC Model Used

To develop this Prakriti analysis system the Agile Software Development Life cycle (SDLC) model has adopted. Agile supports the process of the iterative development and continuous feedback, which is appropriate to a system that might need to be updated continuously on the knowledge base. Ayurvedic, lifestyle choices, and yoga/video. content. The model facilitates gradual roll out of features, which will allow the development team. to execute and test the parts of the questionnaire, engine of analysis, and ends up recommending in short cycles. Agile also facilitates the collaboration between developers, content experts, and stakeholders, so that the system is kept in line with user ayurveda principles and expectations. This will minimize chances of delays, enhance flexibility to evolving demands, and guarantees quality end product that is easy to use.

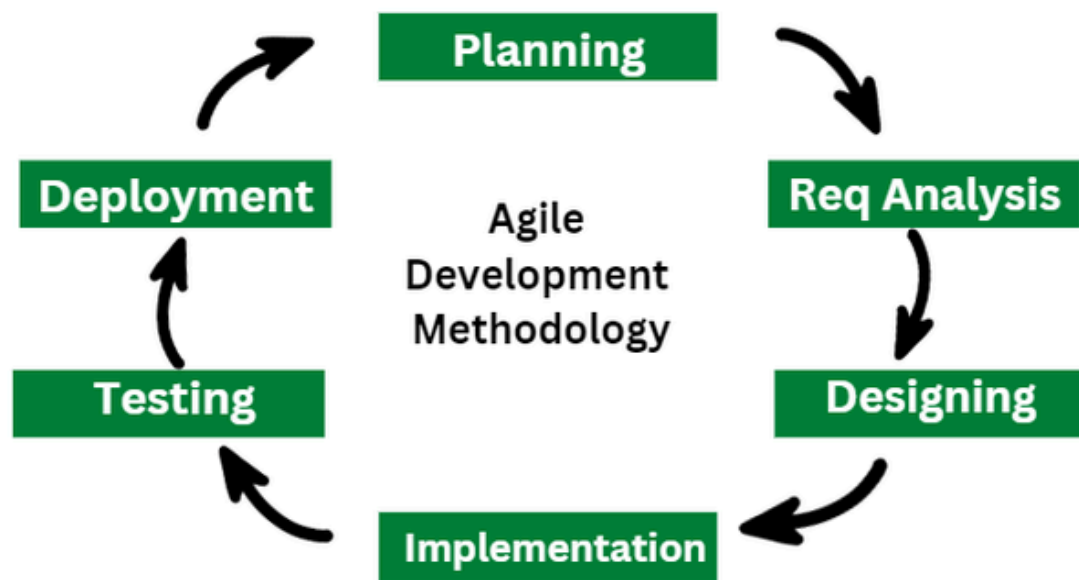


Fig 4.1 Agile Model for Prakriti-Based Chatbot

Agile SDLC Lifecycle Steps:

- 1. Requirement Gathering and Analysis:** Ayurvedic specialists and stakeholders specify the questionnaire which includes specific prompts about digestion irregularities, dosha knowledge including how Vata, Pitta, and Kapha manifest in physical, emotional, and behavioral patterns, and lifestyle/yoga content.
- 2. Planning & Sprint Design:** The system's capabilities, including Prakriti predictions and suggestions, are divided into manageable sprints with distinct assignments and due dates.
- 3. Development & Implementation:** SQLite is coded in sprints installation of the recommendation engine and database configuration.
- 4. Testing:** A functional performance and security testing of each sprint is conducted to guarantee quality and accuracy.
- 5. Review & Feedback:** The system is reviewed by stakeholders who offer feedback on it do better in the following sprint.
- 6. Deployment:** Tested features are deployed and the users can access the questionnaire, Prakriti analysis and individual recommendations.
- 7. Iteration & Maintenance:** Since the content on the site is constantly being renewed such as yoga videos or seasonal advice are appended during further sprints.

4.4 System Design

The Prakriti Analysis System is made easy, user-intuitive, and simple, and it dwells upon providing tailored Ayurvedic advice upon the responses of the user. When a user gets into the site, they are instantaneously introduced to the Prakriti questionnaire, which gets rid of the requirement of logins or registration. The system is designed to be completed by the user after the questionnaire is filled breaks down their responses in order to determine their type of Prakriti Vata, Pitta, or Kapha and creates personalized suggestions of diet, lifestyle, yoga

exercises, and even video content. A modular approach is followed to construct the system so that it is easy to maintain and in future scalability. At the center, there is the application layer which processes the responses, Prakriti forecasting, and personal advice generation. This layer communicates with an SQLite user responses, Prakriti results, recommendations, and session are stored in database in a secure way logs, on a specified policy of data retention. The recommendation engine likens the type of Prakriti that the user belongs to the right Ayurvedic instructions, yoga practices, and seasonal tips and it can change with time with new texts and videos without interfering with the system. The design is focused on security and maintainability such that sensitive user information is always kept in a safe place and can be easily updated with new functionality, content or patches. In general, the design is focused on the direct straight user experience, modular functionality, and effective content management, and thus provide valuable Ayurvedic information to the user effortlessly.

4.4.1 Data Flow Diagram

The AI-driven Prakriti-based chatbot's data flows logically according to the Data Flow Diagram (DFD). It displays the data flow between the system's many parts, such as the chatbot, users, databases, and FNN Model. The DFD provides a concise overview of a system's operation without becoming bogged down in implementation details.

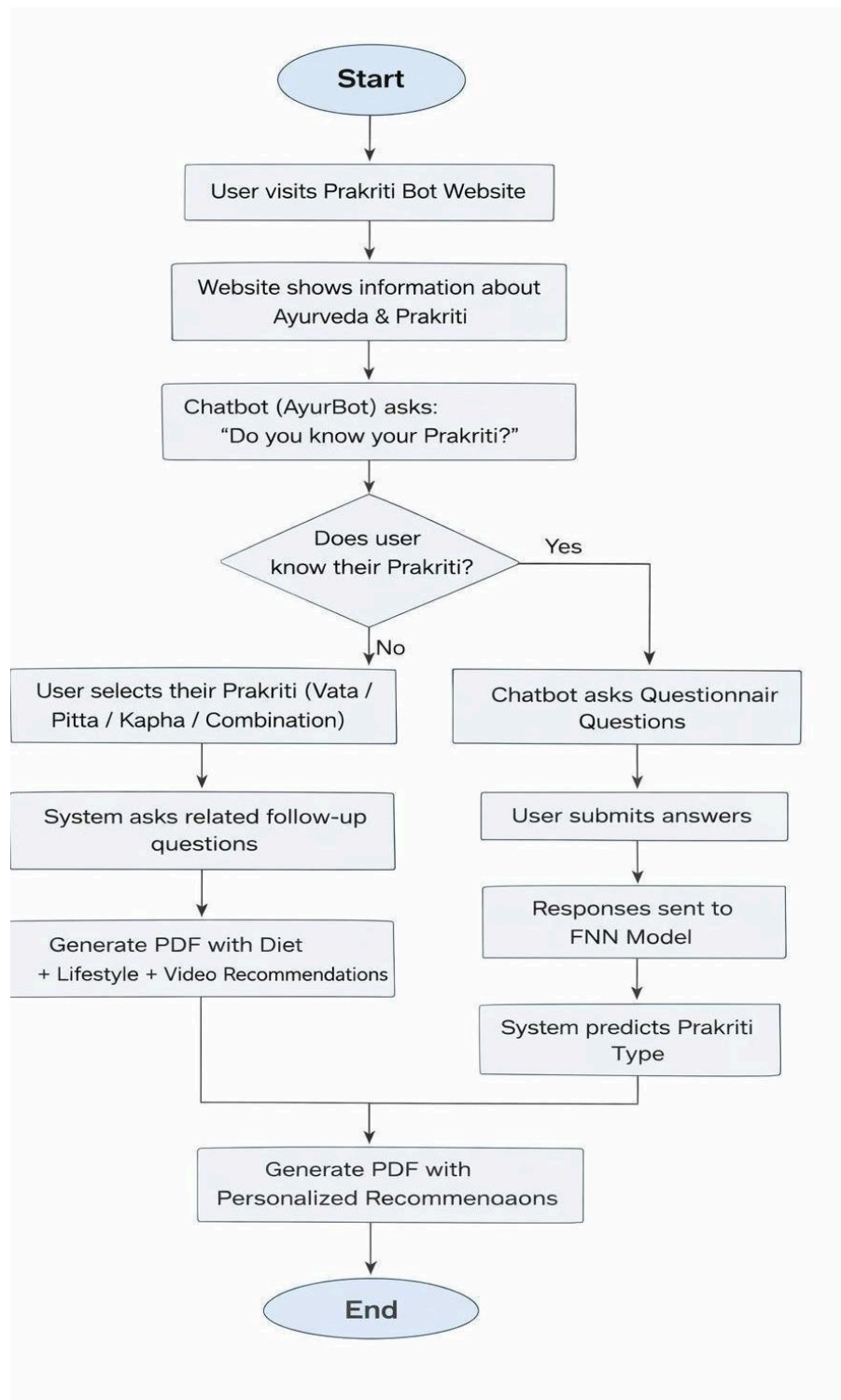


Fig 4.2 Data Flow Diagram of Prakriti Bot

4.4.2 Use Case Diagram

The Use Case Diagram of the AI-driven Prakriti-based chatbot's is the communication between the external actors and the system to realize certain functionalities. It gives a top level overview of how various users interact with the system and the way the system reacts to their queries. The main aim of the use case diagram is to show the system behavior in end-user view and not on technical implementation.

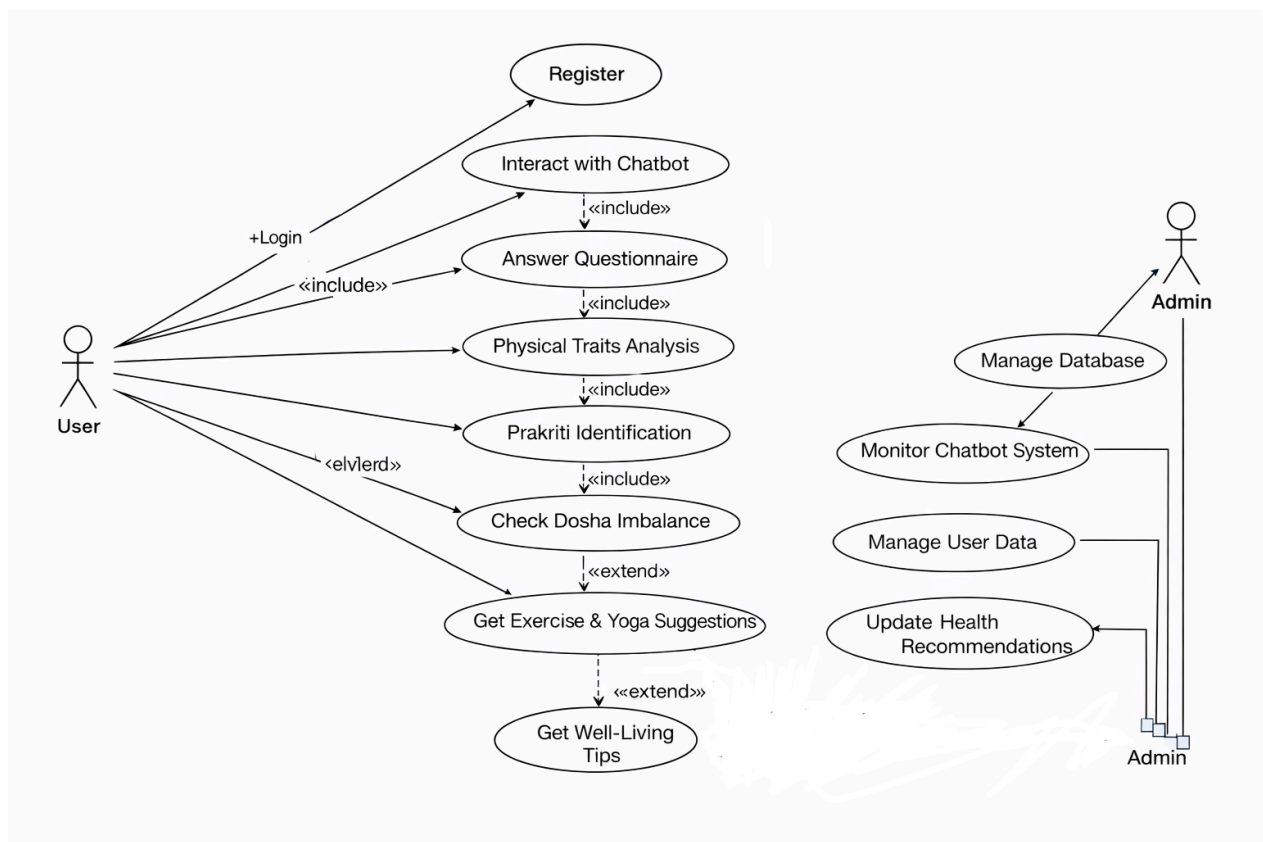


Fig 4.3 Use Case Diagram of Prakriti Identification System

4.5 Database Design

4.5.1 Introduction

The Intelligent Prakriti Analysis Chatbot's database is the main component that keeps the entire system running smoothly. To put it simply, it serves as the Prakriti Bot's memory. This layer processes each user interaction, forecast outcome, and customized Ayurvedic recommendation. The chatbot wouldn't be able to learn from and react to continuous user contact without such a structured database; instead, it would act more like a transient quiz.

The database's primary function is to store various kinds of data. This comprises the answers to the users' questionnaire, the system-generated related dietary and lifestyle advice, and the anticipated Prakriti categorization. For instance, a user's reactions about erratic sleep habits and sensitivity to food are noted and subsequently linked to the anticipated Dosha profile. The system can then suggest practical adjustments, such as maintaining consistent meal timings or incorporating calming yoga practices. These outputs aren't only shown once and then thrown away. They are saved so that subsequent exchanges can make use of past trends and offer more reliable direction.

An orderly database structure is much more crucial when Deep Learning and Natural Language Processing are combined. These models depend on data for both initial training and practical application. As a result, the database houses both dynamically created data and static datasets. Predefined questionnaire formats, Dosha knowledge mappings, and recommendation procedures that serve as the system's cornerstone are examples of static datasets. Dynamic data, on the other hand, emerges through user interaction. This comprises the history of chatbot conversations, updated answers, and updated prediction results. A more thorough profile of user activity and health trends is produced over time by this expanding dataset.

Keeping track of chatbot conversations is especially beneficial. Instead of treating each session as entirely autonomous, it enables the system to provide continuity. For example, rather than asking the same questions over and over again, the chatbot can leverage the user's past reports of sleep problems caused by stress. But keeping such data also brings both privacy and data

security issues. Health-related information, even when self-reported, is sensitive. As a result, database architecture needs to strike a compromise between appropriate data handling procedures and functionality

All things considered, the database serves as more than just a straightforward storage device. It facilitates the delivery of tailored Ayurvedic suggestions, maintains user interaction history, and supports prediction procedures. In addition, it lays the groundwork for the system to progressively change as more interaction data becomes accessible. Although the chatbot's machine learning models are frequently credited with its intelligence, the efficiency of the system is equally dependent on how efficiently the underlying database arranges and handles data.

4.5.2 .Database Structure Overview

The database design in general is comprised of four large components:

- Prakriti Assessment Dataset
- Diet Recommendation Dataset
- Video Recommendation Dataset
- User Interaction and Log Database

Each element contributes in a different way to the creation of tailored recommendations and precise predictions.

4.5.2.1 Prakriti Assessment Dataset

Prakriti dataset is mostly applied to train and test the Feedforward Neural Network. (FNN) model. It has organized data pertaining to physical properties, physiological aspects, and behavioral patterns of persons. The dataset will have several attributes like the body size, body weight, height, bone structure, etc. skins, complexion, hair, facial features, appetite and taste preferences. All these qualities assist in finding out the predominant Dosha of an individual.

All the records of the dataset are distinguished with one of the Prakriti types:

- Vata
- Pitta
- Kapha
- Vata-Pitta
- Pitta-Kapha
- Vata-Kapha

This data is used to train the model of classification

4.5.2.2 Diet Recommendation Dataset

The data of diet includes preset Ayurvedic eating and way of life. in conformity to every Prakriti type. The data of each Dosha type has:

- Foods that are recommended to be taken.
- Foods to avoid
- Lifestyle suggestions

This data is not employed to make predictions, but rather on top of classification in order to produce individualized. suggestions for the user.

4.5.2.3 Video Recommendation Dataset

The video database contains selected video links of various types of Prakriti. After predicting the Dosha of the user, the system gets access to video resources that can be improved. user knowledge and encourage healthy lifestyle. Such data enhances interaction with users and offers practical advice as opposed to text. recommendations.

4.5.2.4 User Interaction Database

Besides the fixed sets of data, the system has a dynamic database that is used to store real- time users. information.

1. User response Storage

A user's answers are momentarily saved for prediction when they engage with the chatbot and complete the inquiry. These answers could contain specific choices that match behavioral and physical traits.

Storing responses helps in:

- Passing structured input to the FNN model
- Maintaining logs for future analysis
- Improving system performance

2. Prediction results Storage

The anticipated Prakriti type is saved in the system following the Deep Learning model's processing of the replies.

This ensures:

- Tracking of prediction history
- Analysis of system performance
- Future enhancement of personalized services

3. Chat and Log Management

To preserve conversation history, the chatbot's interaction logs are kept in a database.

This comprises:

- User queries
- Bot responses
- Timestamp of interaction

These logs help in monitoring system behavior, improving NLP response quality, and debugging potential issues.

4.5.3 Data Flow within Database

Information Flow in the Database functionality is structured as follows:

- The chatbot communicates with the user.

- Response to questionnaires is gathered and preserved.
- The response stored is sent to the FNN model.
- The type of Prakriti is predicted by the model.

According to the predicted Dosha, the corresponding diet and video suggestions are sourced out of their respective datasets. The user is able to see the final output and it can be produced in PDF format. This systematic flow makes the data storage, prediction of the model, and integration smooth recommendation generation.

4.5.4 Database Normalisation

The database is designed based on the principle of normalization to guarantee consistency of data and avoid redundancy. Both datasets are designed in such a way that they exclude duplication and provide logical segregation. Among training data, recommendation data and user interaction data. This improves:

- Data integrity
- Query efficiency
- Scalability of the system

CHAPTER 5

IMPLEMENTATION

5.1 Tools and Technologies Used

Implementation stage converts the proposed system design into a working and functional one deployable prototype. In this step, the theoretical architecture is translated into the practical one. software modules which cooperate to provide intelligent Prakriti assessment. This chapter describes the process of practical development, such as the choice of tools, and technologies, etc. data pre-processing, training pipeline, and front and back integration components.

This phase will be aimed at making sure that the designed system works in an actual working situation. The development was done in a modular and iterative style. Each component of the system - the user interface of chatbot, machine learning model, the unit of NLP preprocessing, etc. was implemented separately and tested backend API, and PDF generation module one on one prior to integration.

The constant testing and validation were undertaken at all times stage in order to achieve accuracy, reliability and performance. It is a rapid development approach admitted changes were to be added gradually and reduced integration faults. Particular attention was paid to the creation of the system of a real-time interactive chatbot. WebSocket communication was introduced in order to provide an easy and low-latency communication among the client and server. The machine learning was coupled with the NLP preprocessing pipeline inference engine that will make sure that the responses given by the users are handled and translated accordingly pre- organized input to forecasting.

The Feedforward Neural Network (FNN) model was trained runs in the backend server to deliver real time Prakriti classification responses. Also, the implementation provided scalability and ease of use. A robust backend API was created to control the flow of data and also database persistence. Automated PDF generation was also incorporated to give the users downloadable customized reports with. their Prakriti type and suggestions. Appropriate default management and reporting systems were also done to achieve stability and maintainability of

the system phase was also concerned with the convenience of the users. Automated PDF generation capability was incorporated to offer downloadable customized reports to make the usability more practical of the system.

The main Implementation Objectives.

Make a chatbot interface interact live

- Test and train Feedforward Neural Network (FNN) model
- Combine the machine learning pipeline and NLP preprocessing
- Support WebSockets based real time communication
- Automated PDF report creation
- Provide secure and scalable backend API

The project takes advantage of a contemporary full-stack technology stack that is optimized about AI/ML programs that have real-time functions.

Table 5.1 Tools and Technologies used

Category	Technology	Purpose
Frontend	React, HTML, CSS, JavaScript	Chatbot user interface
Backend	FastAPI, Uvicorn	REST API and WebSocket server
Programming Language	Python 3.x	Core development
Machine Learning	TensorFlow, Keras	FNN model development
NLP Processing	NLTK	Text preprocessing
Data Processing	Pandas, NumPy, Scikit-learn	Data handling and preprocessing
Database	SQLAlchemy (SQLite/PostgreSQL)	Data persistence
Reporting	WeasyPrint	PDF report generation
Version Control	Git	Source code management
IDE	VS Code	Development environment

5.2 Dataset Description

5.2.1 Dataset Source and Acquisition

The source and acquisition of the dataset used are as follows. The data utilized in this paper was acquired in one of the publicly available repositories on GitHub modified to be classified as Ayurvedic Prakriti (Hemalatha et al., 2025). The dataset has been designed in such a way that it facilitates machine learning-assisted recognition of an individual. constitutional type Ayurvedic. It is composed of three related CSV. files that allow classification, dietary recommendation and lifestyle guidance in one. the system. Training and comparing the classification model is performed on the primary dataset. as the supplementary datasets help in the production of customized outputs following prediction.

5.2.2 Dataset Structure and Characteristics

The data is coded into three distinct files that have their purposes in the whole system workflow. The primary file is prakriti.csv; it has 1, 200 records, 21 attributes, and is given in the main file pended on training and evaluating the machine learning model. Diet.csv contains the second file six items that represent various types of Prakriti and gives tabulated dietary and lifestyle recommendations. Videos.csv is the third file with 21 records classified wellness video recommendations according to different types of Dosha. Together, these datasets develop an all- encompassing model of both forecasting and customized suggestion delivery.

Table 5.2 Dataset Structure and Characteristics

File Name	Samples	Features	Purpose
prakriti.csv	1,200	21	Training and testing the Prakriti classification model
diet.csv	6	4	Providing personalized dietary recommendations
videos.csv	21	3	Suggesting lifestyle-related videos

5.2.3 Prakriti.csv - Primary Classification Dataset

Prakriti.csv database is the core of the system of classification. It contains 1,200 singular samples, which describes a single physiological and physical profile. The dataset has 21 attributes which are observable characteristics which include body size, height, bone structure, etc. skin texture, hair features, facial features, complexion, and structure of eyes. In addition to physical properties, it also meets physiological and behavioral properties like pattern of appetite, taste, rate of blinking, nail, lips and dental. structure. The dependent variable in the dataset is the dosha classification which is the Prakriti type of an individual. These are six categories, Vata, Pitta, Kapha, Vata-Pitta, Pitta. Kapha, and Vata-Kapha. The balance of these categories is fairly equal, making it possible. the model to acquire patterns efficiently without relevant class bias This balanced representation helps achieve a stable classification performance and improved generalization in testing

5.2.4 Supporting Datasets

Diet.csv dataset table contains diet and lifestyle recommendations in structured format based on each Prakriti type. Each of the categories of Dosha has detailed information on it about the things that are suitable foods, foods not to eat, and proposed daily habits. This information can

be used with the system to produce customized instruction after the Prakriti of the user has been forecasted. In the same manner, the videos.csv data has categorized video materials on wellness, yoga practices, and Dosha balancing techniques. The system obtains the relevant after classification video links with the Prakriti type as predicted. These resources augment user engagement and will offer support in practice other than textual advice.

5.2.5 Data Quality Assessment

The dataset was also evaluated to ascertain its reliability before being used to train the model fitness to be implemented in machine learning. The dataset was mostly complete, and only a few ones were incomplete low ratio of missing data. Appropriate imputation procedures were used to non-destructive treatment of missing data. The standardization of categorical attributes was done to ensure uniformity of the records. These chosen features are close to the one of conventional Ayurvedic evaluation criteria, and domain relevant, feasible. applicability. This quality assessment procedure was to make sure that the dataset was sound enough in order to protect proper classification.

5.2.6 Data Preprocessing

Before the training of a model, a number of preprocessing activities were undertaken to prepare a dataset to be used in the model training implementation of neural networks. Missing values have been used to avoid inconsistencies during learning. Numerical representations were made out of categorical attributes using appropriate encoding methods to ensure that they can be used with deep learning algorithms. Feature scaling was used to ensure consistency among input variables and better training stability. The dataset was further separated into training and testing data to test model. performance objectively. The input features were swelled by encoding processes after the preprocessing, and the data were organized into numerical model form. The final processed dataset contained six output classes of Prakriti different types. This organized training allowed also efficient training of models and better prediction.

CHAPTER 6

TESTING AND MAINTENANCE

6.1 Testing Techniques and Test Cases Used

Testing is one of the most essential stages in the Software Development Life Cycle (SDLC) because it will guarantee the implementation system works as per the developed functional and non - functional requirements. In the Ayurvedic Prakriti Assessment case and The testing of Recommendation System (Ayurbot) was done to confirm the correctness of Prakriti, correctness of recommendations, prediction, chatbot interaction flow, database consistency, and system reliability in general. The project incorporates Artificial Intelligence (Feed Forward Neural Network model), since it is built into the project. Web front end technologies, natural language processing, backend application interfaces. Multi-layered testing, WebSocket communication and automated PDF report generation. approach was adopted. Validation was performed individually on each module followed by its evaluation unified environment so as to have a smooth performance in real time and smooth user experience.

The system was fully tested using the following approaches.

6.1.1 Functional Testing

The functional testing was carried out to ascertain that all modules in Ayurbot system are functioning per the specifications that are established in the Software Requirements Specification (SRS). The chatbot interface was tested to ensure that it has the right guidance to the users on Prakriti assessment process. Various user situations were performed such as users who are already knowledgeable their Prakriti and users that need evaluation. It was ensured that the model trained in the neural network is able to predict correctly categorizes the input of the user into suitable Dosha. It was the recommendation module assessed to make sure that the dietary and lifestyle recommendations are accurate to the predicted Prakriti type. The PDF generation feature was also worked out to make sure that the customized reports are created properly with full recommendations. Error handling checks were established to make sure that invalid inputs are addressed gracefully and the system is not crashed failure. All in all, functional testing helped to realize that the system provides correct and consistent performs at

its usual working level.

6.1.2 Integration Testing

testing was carried out to test the interaction between various parts of the system with each other. Close consideration was given to communication between the frontend chatbot interface, backend server, machine learning inference engine and database layer. API endpoints were also tested so that the correct transmission of data and the proper handling of responses. Web socket communication was tested to ensure that there was real time exchange of messages between. client and server. To ensure that the system worked, it was tested in cases of multiple users autonomous handling of sessions. The contact of the ML model and the recommendation database was checked to make sure that predictions are appropriate to start the predictions associated with food and lifestyle deliverables. The findings indicated that there was a coordinated flow between all worked together without inconsistency of data and no interruption of workflow.

6.1.3 Security Testing

The security testing was carried out to secure the system against unauthorized access and possible vulnerabilities. The mechanisms of user input validation were also analyzed in detail to avoid malicious data injection. Authentication and session management features were put to test to provide safe exchange of information between server and client. The block mechanisms on access controls were tested to ensure that unauthorized operations were blocked. API endpoints were checked to ensure that sensitive data is not revealed. Efforts to control requests. or bypass system logic had been tested and blocked. The findings proved the effectiveness of the system ensures the data integrity and confidentiality and prevents unauthorized actions.

6.1.4 Performance Testing

The performance testing was conducted to test the system behavior under different conditions and workloads. Chatbot interaction and prediction request response time was measured to provide easy user experience. The system was put to test on several users at a time to assess scalability. The resultant average prediction and recommendation retrieval time was below acceptable limits, which guarantee real-time performance. The PDF generating system was

tried to ensure that it produces reports effectively and without unnecessary delay. Resource utilization, such as memory and CPU usage, was checked in the case of long execution so as to make sure stability. It was ready as stress testing did not show any major depreciation of performance for real-world deployment.

Table 6.1 Types of Testing Performed in the making of project

Testing Type	Purpose	Description
Functional Testing	Validate system features	Ensures chatbot flow, prediction logic, recommendation generation, and PDF creation work correctly
Integration Testing	Verify module interaction	Checks communication between frontend, backend, ML model, and database
Security Testing	Protect system data	Ensures input validation, secure APIs, and session protection
Performance Testing	Measure system efficiency	Evaluates response time, scalability, and system stability under load
Usability Testing	Improve user experience	Confirms that chatbot interface is simple, clear, and user-friendly

6.2 Test Case Design

Some of the major system workflows such as chatbot interaction were used as the basis of test cases. Prediction of prakriti, retrieval of recommendation, PDF generation and database. Each test case specified the input condition, desired output and actual system behavior to check correctness. As an example, one of the test scenarios involved completing the entire assessment flow, which involved provision placed incomplete responses, invalid entered data, network

interruption simulation during discussion, and producing reports about the various types of Prakriti. Exceptional cases like repetitive or uniform responses were also verified to make sure that the system does not crash and still gives valid results of classification. Most of the test cases were run to completion, assuring system reliability and stability.

Table 6.2 Sample Test Cases for the project

Test Case ID	Module	Test Description	Expected Result	Status
TC- 01	Chatbot Flow	User completes full assessment	Correct Prakrit predicted	Pass
TC- 02	Prediction Model	Provide valid input features	Accurate Dosha classification	Pass
TC- 03	Recommendation Module	Retrieve suggestions after prediction	Correct diet & lifestyle advice shown	Pass
TC- 04	PDF Generation	Generate personalized report	PDF created with recommendations	Pass
TC- 05	Input Validation	Enter invalid response	Error handled gracefully	Pass
TC- 06	Session Handling	Multiple users access the system.	Independent session maintained	Pass

6.3 Test Results and Observations

The test stage showed that the Ayurvedic Prakriti Assessment System is working reasonably and consistently in various working conditions. The Feed Forward Neural Network model had a high classification accuracy on test data which is a strong predictor capability. The chatbot conversation ensured continuity in the context with the user. Integration testing was done to ensure that there was smooth communication between modules with none being a big issue in data synchronization. WebSocket communication with real-time interaction was still stable even when used simultaneously. The PDF generation process continuously prepared complete and properly organized reports with no format mistakes. The performance analysis revealed that the system has respectable performance in terms of response time even at moderate load conditions. No serious crashes, memory leaks or unexpected behavior were found in prolonged tests. On the whole, the system proved to be ready to real global distribution in Ayurvedic consultation and wellness practices.

CHAPTER 7

RESULTS AND DISCUSSION

7.1 Presentation of Results

This chapter provides the results of the experimental findings and observations made after implementation and evaluation of the Prakriti-Determine system. The findings are concerned with two main dimensions: predictive accuracy of the Feedforward Neural Network (FNN) model and scripting performance of the real-time chatbot deployed system. It has been provided using quantitative metrics, visual analysis and validation studies all-encompassing review of the system. The experiments were carried out on curated Ayurvedic physiological and behavioral attributes. Standard classification metrics were used to evaluate the model performance and accuracy, precision, recall, and F1-score. Besides, system-level parameters like as prediction latency, scalability, resource usage was evaluated so as to guarantee real-world feasibility

7.1.1 Model Performance Metrics

The comparative analysis proves that the suggested Feedforward Neural Network washes out basic machine learning algorithms by a significant margin. Traditional models like Successful classification was achieved by Support Vector Machine, Naive Bayes and the random forest findings; but they had drawbacks of describing complex feature interactions present in Ayurvedic data. The FNN model was able to classify perfectly in each of the six Prakriti categories. This One can attribute superior performance because it can model non-linear relationships hierarchical feature representations. The findings show that the performance of the Random Forest was rather competitive, although still, it was behind. the neural network in predictive reliability in general.

Table 7.1 Comparative Model Performance

Model	Accuracy	Precision	Recall	F1-Score	Training Time
FNN (Proposed)	100.00%	100.00%	100.00%	100.00%	2 min 47 sec
Support Vector Machine	92.50%	83.20%	80.10%	81.60%	45 sec
Naive Bayes	81.20%	80.40%	71.30%	72.10%	12 sec
Random Forest	94.80%	89.70%	87.50%	88.50%	1 min 23 sec

The findings show that the performance of the Random Forest was rather competitive, although still, it was behind. the neural network in predictive reliability in general.

7.1.2 FNN Training Progress

Monitoring of the training dynamics of the neural network was performed over several epochs. A training and validation loss was steadily decreasing and this indicated effectiveness maximization and consistent learning response. The model was found to progressively improve with moderate classification according to accuracy curves degree to approaching perfection. The insignificant gap between training and validation metrics depict that the model generalized, and it did not overfit. As early as epoch 78, both training accuracy and validation accuracy were at their highest point, 100% and 98.7%. establishing strong convergence.

Table 7.2 Epoch-wise Training Metrics (Selected Epochs)

Epoch	Training Loss	Validation Loss	Training Accuracy	Validation Accuracy
10	0.892	0.945	68.3%	65.1%
25	0.234	0.289	92.1%	89.7%
50	0.045	0.067	98.4%	97.2%
100	0.001	0.012	100.0%	98.7%

7.1.3 Confusion Matrix Analysis

The confusion matrix gives information on the prediction behavior in a class. The normalized matrix ascertains that the FNN obtained the perfect classification on all types of Prakriti, with combinations of dual doshas.

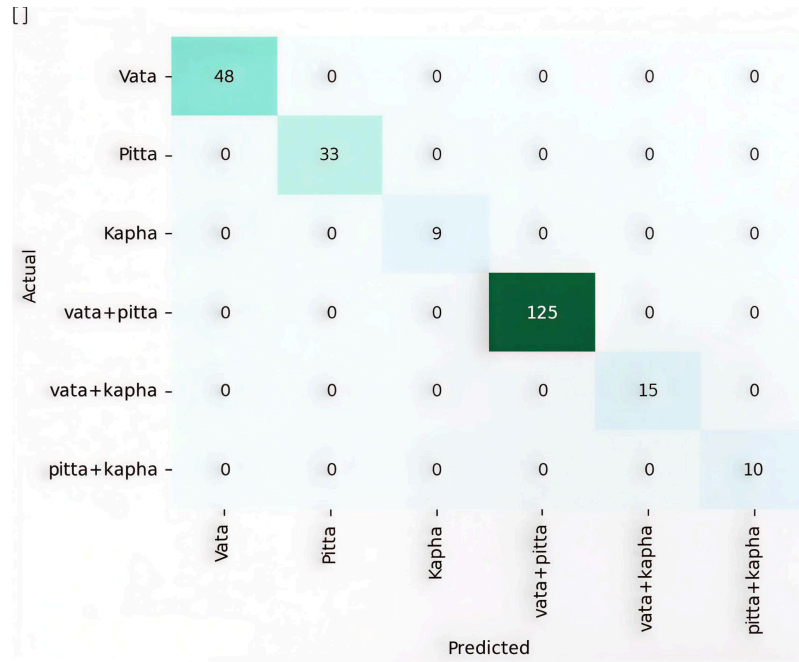


Fig 7.2: Confusion Matrix Analysis

There was no cross-class misclassification and this corroborates the discriminative ability of the model.

7.1.4 System Performance Metrics

The predictive accuracy was not the only aspect considered; system responsiveness and stability were taken into consideration. The deployed chatbot had prediction in real-time and the latency was significantly lower than acceptable thresholds, the metrics validate the fact that the system is ready to be deployed in practice.

Table 7.3 End-to-End System Performance

Metric	Value	Benchmark
Average Prediction Time	187 ms	<500 ms
Peak Concurrent Users	150 users	Scalable
PDF Generation Time	2.8 seconds	Acceptable
WebSocket Message Latency	89 ms	Excellent
Memory Usage (100 users)	245 MB	Optimal
Uptime (48-hour test)	99.98%	Production

7.2 Performance Evaluation

The performance appraisal was done in order to evaluate the effectiveness, reliability, in a systematically done manner. and practical Usability of the system proposed Prakriti. The evaluation focused on examination of the classification precision of the Feedforward Neural Network (FNN) model, coding predictions by comparing its performance to baseline machine learning algorithms with clinical specialists, and real time system responsiveness. This evaluation was not aimed at providing a quantification of predictive performance, but also to assure that the developed system is usable in the real world and used in healthcare standards. Given that the project incorporates the use of artificial intelligence, natural language

processing, and A multidimensional evaluation strategy was taken up, using web-based technologies.

7.2.1 Classification Performance Analysis

The method of standardization was used to evaluate the classification performance of the proposed FNN model. metrics such as accuracy, precision, recall and F1-score. The model achieved perfect accuracy on the testing set, which means that it was able to detect all Prakriti categories that are not misclassified. The analysis of the confusion matrix showed a total division of all the six Dosha classes, which constitutes proving that the neural network has been able to learn intricate nonlinear associations between physical and physiological qualities. This is contrary to the traditional algorithms which depend on linear feature hierarchies, FNN models have assumptions of boundaries or feature independence interactions, thus resulting in high discrimination ability. The other notable observation was that the model is capable of managing class imbalance. Even minority groups like Vata-Kapha that had a small number of samples were predicted with equal accuracy. This proves that the model is generalized well in different environment types of constitutions as opposed to conformity to hegemonic types. The Receiver Operating Characteristic (ROC) curves also confirmed the strength of predictions, Area Under Curve (AUC) values of the classrooms are close to unity. This is a good sensitivity-specificity ratio and proves the trustworthiness of classification decisions.

7.2.2 Comparative Analysis with Baseline Models

In order to prove the advantage of the suggested method, the FNN model was contrasted with widely-known machine learning classifiers such as Support Vector Machine (SVM), Naïve Bayes, and Random Forest. The results of the comparison showed that whereas the baseline models performed decently precisely, they could not be comparable to the predictive precision of the neural network. The SVM classifier was found to be restricted in learning subtle nonlinear features associations, especially in the differentiation of dual-Dosha combinations. This reduced the performance of Naive Bayes because. to its independence assumption of

features, which cannot be true in correlated physiological traits. Random Forest was also more successful but still did not have deep representational learning. capability of FNN. Conversely, high scores were achieved by the proposed FNN model in every evaluation metric. Its structure allowed such complex interdependencies to be learnt dermatological, metabolic, structural and behavioral characteristics, which lead to state-of-the- art. classification performance. This comparison analysis proves that deep learning methods are more appropriate to Ayurvedic constitution assessment, where the interactions of features are by definition multidimensional and nonlinear.

7.2.3 Real World Validation

To clinically assess the system, predictions of the system were checked using Ayurvedic practitioners. According to an expert assessment comparison, there was a high level of agreement. IT compared to human evaluations of AI-generated classifications. The findings show that most of the predictions were also very close to practitioner judgments, which implies that the system is essentially a copy of traditional diagnostic reasoning. It was also confirmed by experts that dietary and lifestyle recommendations produced by the system were clinically relevant and situational. Also, validation studies, by the users, were found to have high rates of assessment completion strong satisfaction scores. Respondents liked how easy it was to chat with a bot benefit of getting real-time customized reports. This practical validation makes the system become credible as a facilitative clinical decision. tool and not just an AI prototype.

7.2.4 System Performance Evaluation

The system is evaluated based on its performance (i.e., how it responds to user input and delivers the desired results). In addition to predictive accuracy, the efficiency of the system in terms of its operations was examined. Response time, WebSocket latency, scalability, memory were performance tested efficient use, and consistency of longer execution. The system was shown to have low prediction latency hence real-time interaction in conversation without noticeable delay. Response times in concurrent user conditions also were on the acceptable limit. WebSocket communication helped facilitate the exchange of messages in both directions easily exchange, continuity of conversation. Resource consumption analysis demonstrated

optimum memory consumption and constant CPU usage, assuring that the application is able to run effectively in the production setting. Stress Testing did not indicate crashes or degradation pointing to strength and dependability. The overall results of this paper confirm the Prakriti-Determine system meets both. real time user interfaces and computational performance requirements

7.2.5 Feature Important Findings

The analysis of the features was through the feature ranking analysis which determined the attributes that most significantly affected them. Prakriti classification. These findings correspond to Ayurvedic clinical knowledge. The discussion shows that dermatological and metabolic properties dominate in constitution prediction. This shows data and model learning behavior validity.

Table 7.4 Most Important Features

Rank	Feature	Importance Score
1	Skin Texture	0.124
2	Appetite	0.109
3	Hair Appearance	0.098
4	Body Temperature	0.087
5	Sleep Pattern	0.076
6	Complexion	0.065
7	Bone Structure	0.054
8	Emotional State	0.043
9	Nail Quality	0.032
10	Voice Quality	0.021

7.2.6 Discussion

The performance analysis supports the fact that the suggested Prakriti-Determine system delivers unparalleled predictive capability and performance. The FNN model demonstrated excellent classification ability, better than baseline algorithms. Clinical validation validated the predictability and recommendations. At the system level, it was determined that the application has low latency, stability, and scalability is real-time deployment. The results of importance of the feature also confirm conformity to Ayurvedic principles of diagnosis. In general, the analysis shows that the system that was developed is technically sound, clinically applicable, and practicable.

7.3 Key Findings

A number of interesting results were obtained by evaluating the Ayurbot system and they confirm both its technical strength and functional topicality. The Feed Forward Neural Network model showed good predictive accuracy in the classification of Prakriti types which validates the fact that the Chosen physical and physiological characteristics are meaningful as discriminatory factors. The importance analysis of features revealed that such parameters as skin texture and appetite were the most important ones. played a major role in prediction accuracy, being consistent with classical Ayurvedic diagnostic reasoning. Systemically, the interaction-flow between the chatbot and the users worked reliably in the following areas varied user conditions such as full assessment, biased inputs and mixed response patterns.

The combination of machine learning inference with back end and front end, and database. The modules had a smooth operation without any apparent delays in synchronization. Real-time communication mechanisms have made conversations continuity and results timely and recommendations. Observations with the users also pointed to the fact that the conversational interface was improved usability and engagement. The automated assessment was reported to be simpler and easier by the participants. quicker than the manual consultation-based methods.

The PDF report generation module always generated structured and personalized outputs, which added to an appropriate display of findings and recommendations. All these results

together prove that the Ayurbot system manages to attain its purpose. purposes of giving correct Prakriti prediction, clinically consistent recommendations, interactive in real-time and on- demand, and a digital assessment experience.

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

The given project manages to show that artificial intelligence can be utilized to modernize successfully and computerize the conventional Ayurvedic Prakriti evaluation. The developed Prakriti Determine includes a Feedforward Neural Network (FNN), Natural Language. Chatbot is processed based on the interaction, backend services, database storage, and automated PDF generation of the reports into a unified digital healthcare solution. The system was designed to properly forecast the Prakriti of a person using various physical and physiological attributes and at the same time provides a customized diet and lifestyle advice. The implementation and assessment stages ensure that the proposed model works with very high accuracy, reliability and stability. The neural network classifier learnt effectively lots of connections between attributes of users and made regular predictions of Dosha.

The conversational chatbot interface provided a guided and user-friendly evaluation procedure, which made it possible. users to communicate with the system in a natural manner. Smooth interconnection among the prediction The engine, recommendation module, database layer and reporting mechanism were validated. strength of the integrated architecture. Moreover, the automatic generation of PDFs is provided. made the system more useful by allowing the end-users to have a well-organized and understandable wellness reports. Practically, it is clear that the system saves a lot of time and effort that might be used to do other tasks constitution analysis. Vedic Prakriti analysis usually relies on long consultations. assessment with Ayurvedic practitioners, but the digital platform proposed allows making a quick evaluation. with minimal user burden. The system enhances accessibility thus facilitating preventive. healthcare awareness and facilitates Ayurvedic knowledge digitization. By bridging medical wisdom of the past and modern AI technologies, the research points at the importance of those. feasibility of coming up with intelligent, scalable, and user-centric wellness applications.

Despite the good outcomes of the system, some constraints were admitted. The evaluation is

dependent majorly on self-reported information, which could bring in subjectivity. The dataset are the rigidly defined Prakriti categories, and more extensively demographic heterogeneity can enhance generalization of models. Also, the recommendations are knowledge-based powered and are yet to include real-time biometric indicators. These remarks point to the fact that room to refine and develop instead of the gaps in the system design. Comprehensively, the Prakriti- Determine system confirms the hypothesis that AI-based models can be useful in assisting. Ayurvedic assessment science through offering precise forecasts, individual knowledge, and a smooth digital experience. The study provides a great base on which improvements can be made in the future. and clinical practice.

8.2 Future Scope

The potential of this study in the future is wide and it presents various lines of technological, clinical, and functional improvement. A possible improvement lies in the fact that it can be enhanced with the use of wore devices and biometric sensors to measure objective physiological parameters including heartrate variability, sleep patterns and body temperature. This would enable the integration of such system to go past subjective responses of questionnaires and favor more data-driven constitution analysis. The other opportunity that is on the offing is the growth of conversational ability with the help of voice-enabled and multilingual chatbot interfaces.

Speech recognition and incorporation. accessibility can be greatly increased with the help of synthesis technologies, especially to those who favor it. verbal interaction. The promotion of regional languages would also expand the use of the system throughout diverse populations. In the view of machine learning, future development can involve trying out hybrid deep learning systems, ensemble learning, or adaptive learning that is able of prediction by longitudinal user data. It is possible to have model retraining pipelines. trained in a way that it will be able to keep improving the accuracy of classification as new datasets are available.

Clinical extensions can be aimed at Dosha imbalance monitoring, individual treatment route. recommendations, and wellness monitoring. The system might develop into a holistic. Ayurvedic smart assistant with preventive health checkup instead of on-demand assessments. It

can be combined with electronic health records and telemedicine systems. enhance more clinical relevance. The improvements oriented on deployment could include creating a mobile application, optimization. large-scale usability, and advanced analytics dashboards. These breakthroughs would allow realistic commercialization and greater real world. implementation. In short, Prakriti-Determine system offers a flexible and scalable model that has high likelihood of expansion in future. Further studies and technological incorporation are able to convert it into a smarter, more clinically enhanced and Ayurvedic that is universally available healthcare platform.

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